

E rate is affine in gamma frequency

nb041 · 2 June 2026

The trained networks this entry uses are produced once in the shared training hub, nb022 (Training), and reused here rather than retrained.

Abstract

nb037 varied τ_{GABA} at inference and the per-cell E rate tracked the gamma cycle. Does that survive *re-training* at each τ_{GABA} ? Yes. Across $\tau_{\text{GABA}} \in \{4.5, 6, 9, 12, 18, 27\}$ ms $\times$ 3 seeds, $r_E = 0.87 + 0.195 \cdot f_\gamma$ with $R^2 = 0.996$. The slope is per-cycle E-cell participation; the intercept is a non-rhythmic baseline; accuracy stays at 80–90% across the sweep. The cycle clock constrains what the network can become.

Method

Sweep. Six τ_{GABA} values $\{4.5, 6, 9, 12, 18, 27\}$ ms $\times$ three seeds = 18 networks. The recipe matches the nb025 PING baseline (100 epochs medium tier on MNIST, Adam at 4×10^{-4} , batch 256, mem-mean readout, θ_u off, $\Delta t = 0.1$ ms, $T = 200$ ms); only τ_{GABA} varies.

Measuring f_γ . For each cell, f_γ is the parabolic-interpolated peak of the Welch PSD on the per-trial population E trace; the fit (Figure 4) uses per-trial peak medians, which avoid the centroid bias of trial-mean PSD peaks. Parabolic interpolation with peak-bin values (y_0, y_1, y_2) :

$$f_\gamma = \text{freq}[\text{peak}] + \frac{1}{2} \frac{y_0 - y_2}{y_0 - 2y_1 + y_2} \cdot \Delta f, \quad \Delta f = 5 \text{ Hz}.$$

It is needed because the bare 5 Hz bin quantisation would coarsen f_γ across the six conditions; on a well-isolated peak the interpolation error is $O((\Delta f)^3)$.

The predicted law. The shape $r_E = a + p \cdot f_\gamma$ is predicted by cycle dynamics, not curve-fitted. Within one cycle of duration $1/f_\gamma$, a fraction p of E cells emits exactly one spike (those nearest threshold when the I shunt drops); the rest are still recovering, so the cyclic per-cell rate is $p \cdot f_\gamma$. At long τ_{GABA} the I conductance never fully decays and the cycle dissolves into a tonic bath, leaving a feedforward baseline a independent of f_γ :

$$r_E = \underbrace{a}_{\text{feedforward baseline}} + \underbrace{p \cdot f_\gamma}_{\text{cyclic contribution}}.$$

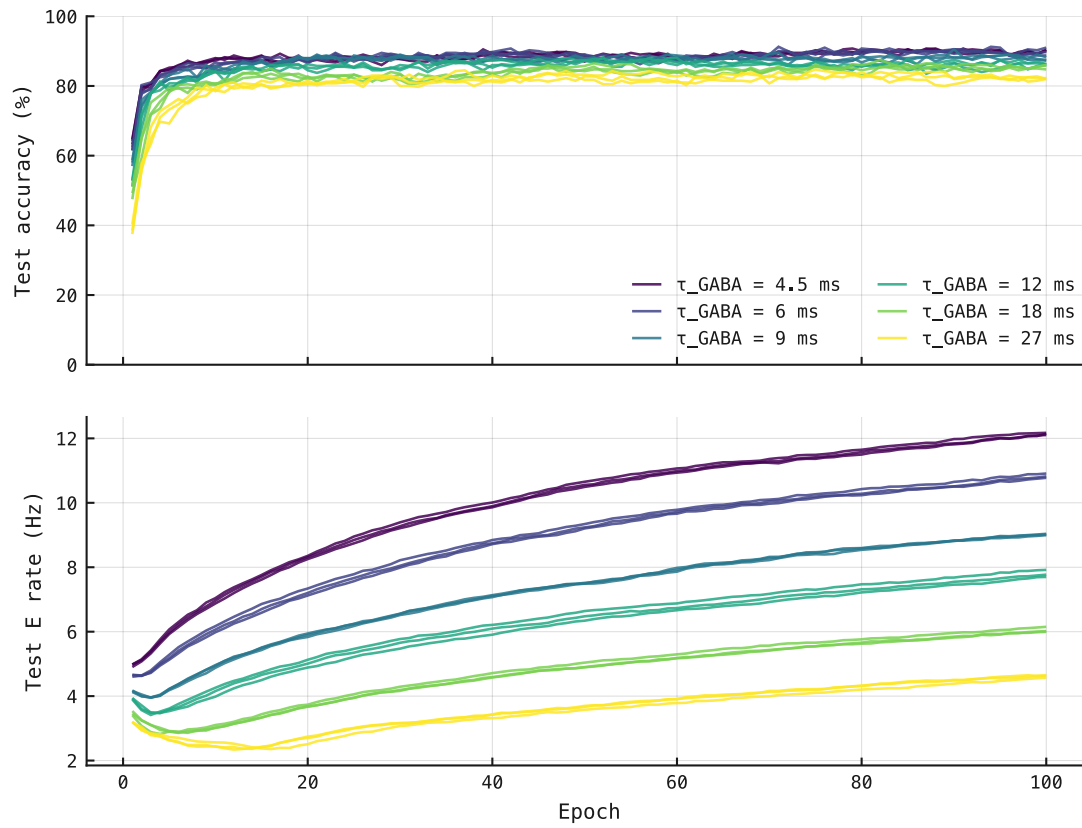
We fit this across the 18 cells.

Convergence. Accuracy plateaus by epoch 15 while the E rate keeps climbing to epoch 70–100 (Figure 1), so the 100-epoch numbers are the converged ones. The fit shape is robust to training length — p tightens with it, and the 100-epoch row is canonical:

training	a (Hz)	p (Hz/Hz)	R^2
30 epochs	1.14	0.166	0.990
100 epochs	0.87	0.195	0.996

Results

Per-cell training curves – convergence check across τ_{GABA} sweep



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Figure 115: Per-cell accuracy (top) and E rate (bottom) over training. Accuracy plateaus by epoch 15; the E rate keeps climbing until epoch 70–100, so the 100-epoch numbers are converged.

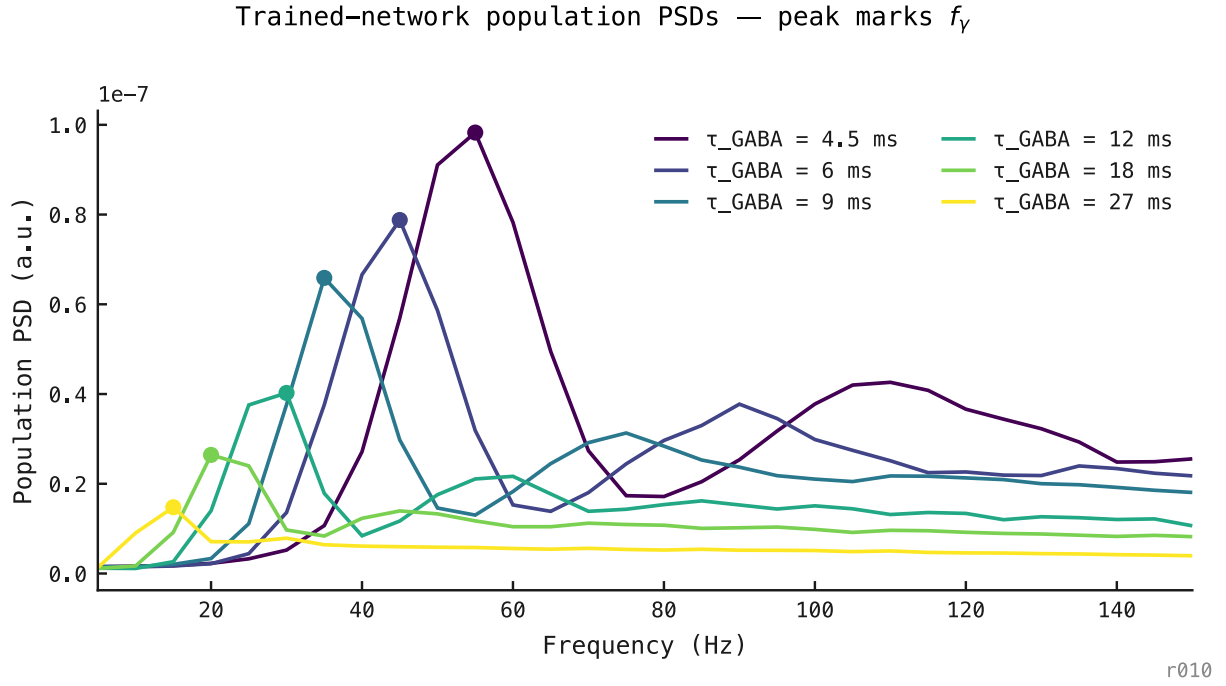


Figure 116: Trial-mean Welch PSDs by τ_{GABA} ; dots mark the parabolic-interpolated peak. The peak shifts cleanly from ≈ 14 Hz at $\tau_{\text{GABA}} = 27$ ms to ≈ 54 Hz at $\tau_{\text{GABA}} = 4.5$ ms, with no overlap between adjacent conditions.

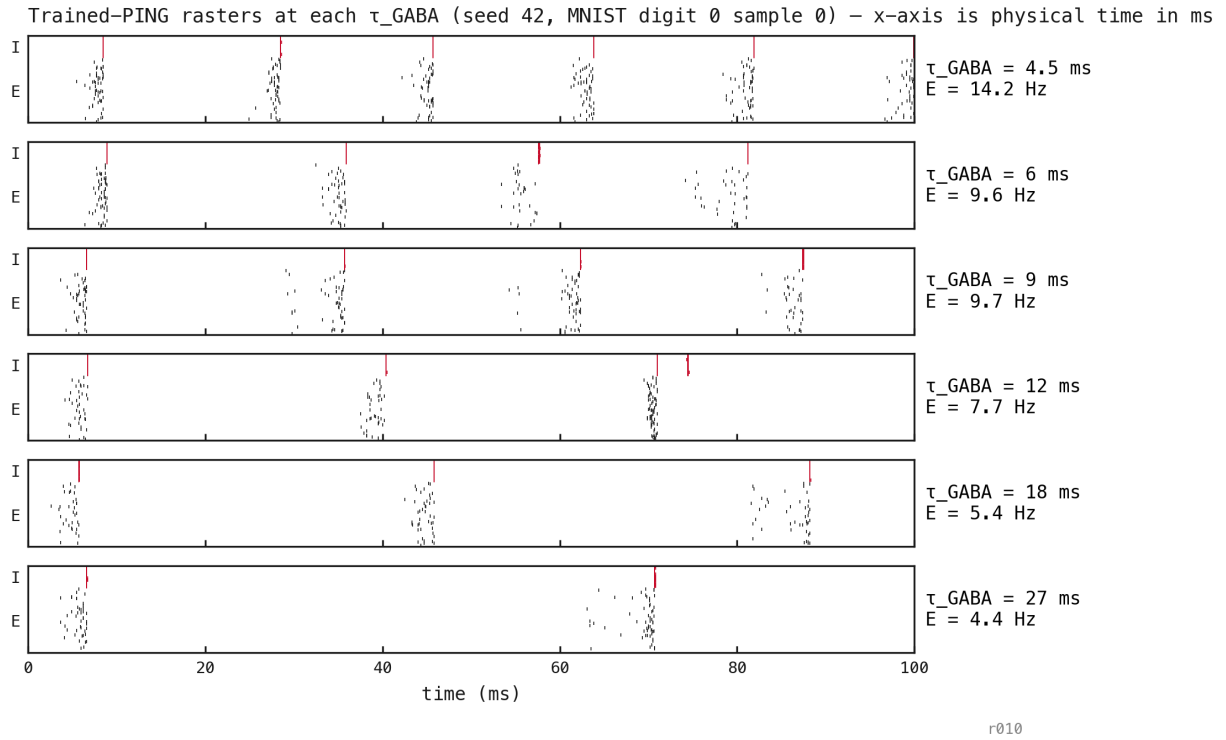
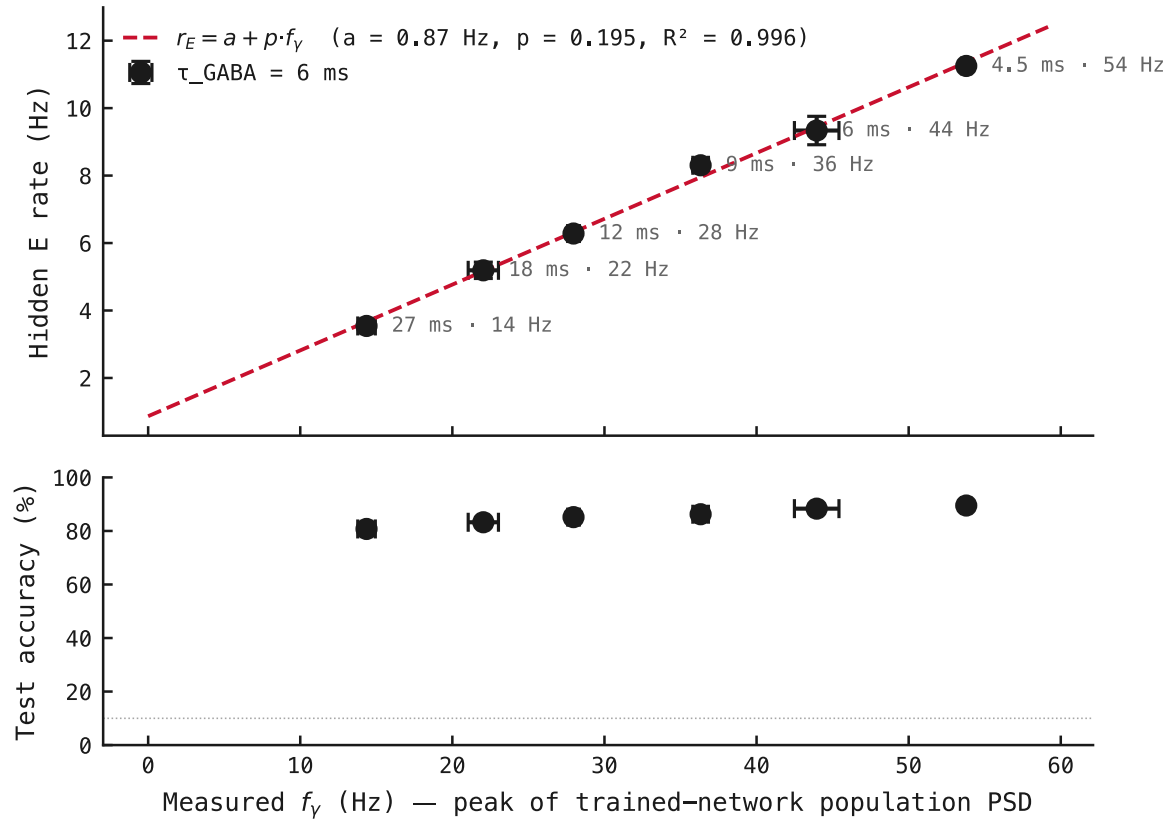


Figure 117: One MNIST trial through each network. The cycle period stretches from ≈ 17 ms ($r_E = 14.5$ Hz) to ≈ 63 ms ($r_E = 4.6$ Hz) — the eye and the spectrum agree.

Retrained τ_{GABA} sweep — post-training E rate tracks measured f_γ



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Figure 118: The law itself. Top: mean post-training E rate vs f_γ , six clusters $\times$ three seeds, error bars from seed variance — the affine fit passes through every error bar. Bottom: per-cluster accuracy is flat, so the rate change is not paid in classification.